

2 Logic

2.1 Statements

Decide whether or not the following are statements. In the case of a statement, say if it is true or false, if possible.

2. Every even integer is a real number.

Statement, true.

4. Sets $\mathbb{Z}$ and $\mathbb{N}$.

Not a statement.

6. Some sets are finite.

Statement, true.

8. $\mathbb{N} \notin \mathcal{P}(\mathbb{N})$

Statement, false.

10. $(\mathbb{R} \times \mathbb{N}) \cap (\mathbb{N} \times \mathbb{R}) = \mathbb{N} \times \mathbb{N}$.

Statement, true.

12. If the integer x is a multiple of 7, then it is divisible by 7.

Statement, true.

14. Call me Ishmael.

Not a statement.

2.2 And, Or, Not

Express each statement or open sentence in one of the forms $P \wedge Q$, $P \vee Q$, or $\neg P$. Be sure to also state exactly what statements P and Q stand for.

2. The matrix A is not invertible.

$\neg P$, where P : Matrix A is invertible.

4. $x < y$.

$\neg P$, where P : $x \geq y$.

6. There is a quiz scheduled for Wednesday or Friday.

$P \vee Q$, where P : A quiz is scheduled for Wednesday, and Q : A quiz is scheduled for Friday.

8. At least one of the numbers x and y equals 0.

$P \vee Q$, where P : $x = 0$, and Q : $y = 0$.

10. $x \in A \cup B$.

$P \vee Q$, where P : $x \in A$, and Q : $x \in B$.

12. Happy families are all alike, but each unhappy family is unhappy in its own way. (*Leo Tolstoy, Anna Karenina*)

$P \wedge Q$, where P : Happy families are all alike, and Q : Each unhappy family is unhappy in its own way.

14. A man should look for what is, and not for what he thinks should be. (*Albert Einstein*)

$P \wedge \neg Q$, where P : Man should look for what is, and Q : Man should look for what he thinks should be.

2.3 Conditional Statements

Without changing their meanings, convert each of the following sentences into a sentence having the form “*If P , then Q* ”.

- 2. For a function to be continuous, it is sufficient that it is differentiable.**

If a function is differentiable, then the function is continuous.

- 4. A function is rational if it is a polynomial.**

If a function is a polynomial, then the function is rational.

- 6. Whenever a surface has only one side, it is non-orientable.**

If a surface has only one side, then the surface is non-orientable.

- 8. A geometric series with ratio r converges if $|r| < 1$.**

If a geometric series has a ratio with an absolute value less than 1, then the geometric series converges.

- 10. The discriminant is negative only if the quadratic equation has no real solutions.**

If a quadratic equation has a discriminant that is negative, then the quadratic equation has no real solutions.

- 12. People will generally accept facts as truth only if the facts agree with what they already believe. (*Andy Rooney*)**

If people accept facts as truth, then the facts agree with what they already believe.